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Limited evidence that body size shrinking and shape-shifting alleviate thermoregulatory pressures in a warmer world



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Amassing evidence indicates that vertebrates across the globe are shrinking and changing shape concurrent with rising temperatures. Ecogeographical theories assert that these changes should provide thermoregulatory benefits by easing heat dissipation, however, thermophysical models underpinning such theories are highly simplified and lack empirical validation. Using data from three temperature-manipulation experiments, we quantified the contributions of body size and appendage lengths toward thermoregulatory performance in Japanese quail, while simultaneously querying neutral plasticity as an alternative driver of avian shape-shifts. In the cold, body mass and leg length (here, tarsus length) influenced energy costs of warming, but only among juveniles. In the warmth, smaller body sizes, longer legs and longer bills independently reduced energy and water costs of cooling across ages, but whole-body phenotypes necessary to provide even moderate thermoregulatory benefits were rare (2.5%) and required large departures from allometry. Last, rearing in the warmth reduced body sizes and increased appendage lengths comparable to recent changes observed in nature, but emergent morphologies provided no clear thermoregulatory benefit. Our findings question whether shrinking and shape-shifting are indeed easing thermoregulation in birds or reflect selection for such. Neutral plasticity, or relaxed selection against small body size in juveniles, may better explain recent avian shape-shifting.

In 2022, ‘Dippy’, the cast replica of a *Diplodocus carnegii* skeleton unearthed in the late 1800s, returned for display at the United Kingdom’s Natural History Museum after several years of absence. Within 6 months of exposure, over 1 million attendants sought to view the cast, rendering Dippy the museums ‘most popular exhibition’ of the year¹. That Dippy stands an impressive 6 metres tall and 26 metres long is undoubtedly one of the keys to its allure; we are fascinated with large-bodied animals^{2,3}. Yet opposing this fascination, amassing evidence indicates that a wide range of extant bird and mammal species are shrinking and changing shape^{4–7}. How or why these ‘shape-shifts’ have occurred is not yet known, however, given a concurrence with rising global temperatures, many have interpreted them as beneficial for thermoregulation^{6,8,9} and potentially driven by selection or adaptive

plasticity. By decreasing body size and adjusting shape, surface area to volume ratios may increase, thus easing heat dissipation in a warmer world.

The logic behind thermoregulation as a driver for species’ shape-shifts is not novel. Over a century ago, Bergmann¹⁰ and Allen¹¹ applied the same logic to explain their observations that body size generally decreases, and appendage length increases, with increasing environmental temperatures across the globe (phenomena known as Bergmann’s and Allen’s rules, respectively). Despite this, whether body size and shape do have meaningful impacts on costs of thermoregulation in endotherms with evolved heat-retention and dissipation mechanisms is tenuous. Already 70 years ago, for example, Scholander¹² argued that low surface area to volume ratios should be irrelevant for endotherm survival in the cold since heat loss is so readily

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